

# PHILIPP BENJAMIN ARETZ

Center for Theoretical Physics, Massachusetts Institute of Technology  
Cambridge, MA, USA



## RESEARCH INTERESTS

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Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

## EDUCATION

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**PhD in Physics**, Massachusetts Institute of Technology, Cambridge, MA, USA 2024–present

**MASt in Mathematics (Part III)**, University of Cambridge, Cambridge, UK 2023–2024  
Pass with Distinction, 92% — ranked 9th of 294

**BSc in Physics**, Heidelberg University, Heidelberg, Germany 2020–2023  
1.0 (best possible)

Thesis: *The Keldysh Functional Integral and Bounds on Truncated Expectation Values* (advisor: Prof. Manfred Salmhofer)

**Abitur**, Goethe-Schule Bochum, Bochum, Germany 2020  
1.0

## RESEARCH POSITIONS

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**Research Assistant**, Heidelberg University, Heidelberg, Germany 2022–2023  
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics  
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.

**Tutor**, Heidelberg University, Heidelberg, Germany 2021–2022

## PUBLICATIONS AND PREPRINTS

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### Published

**P. B. Aretz**, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].

P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

### Preprints

**P. B. Aretz**, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive  $pp$  scattering*, arXiv:2607.07788 [hep-ph].

## TALKS

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### Conference Talks

*Diffractive Across Colliders*, TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder 2026

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|---------------------------------------------------------------------------------------------------|------|
| <i>Diffraction Across Colliders</i> , SCET 2026, Seoul, South Korea                               | 2026 |
| <i>Diffraction at the LHC</i> , GGI School on Theory of Fundamental Interactions, Florence, Italy | 2026 |

#### Posters

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| <i>Diffraction at the LHC</i> , ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna                        | 2026 |
| <i>Generalised Detectors in Heavy Ion Collisions</i> , ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna | 2026 |

#### Internal seminars

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| Diffraction across Colliders, MIT Graduate Student Seminar (2026)                                  |  |
| EFT for Collider Physics, MIT Graduate Student Seminar (2025)                                      |  |
| Generalised Detector Observables in the Heavy Ion Environment, MIT EFT Group Meeting (2025)        |  |
| Resurgence and Renormalons in QCD, MIT EFT Group Meeting (2025)                                    |  |
| Generalised Detector Observables in the Heavy Ion Environment, MIT Heavy Ions Group Meeting (2025) |  |

#### CONFERENCES AND SCHOOLS ATTENDED

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| ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Erwin Schrödinger Institute - Vienna (Poster) | 2026 |
| TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder, CO, USA                                 | 2026 |
| SCET 2026, Seoul, South Korea (Talk)                                                                                | 2026 |
| GGI School on Theory of Fundamental Interactions, Florence, Italy                                                   | 2026 |
| SCET 2025, Tucson, AZ, USA                                                                                          | 2025 |

#### AWARDS AND FUNDING

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| <b>Jennings Prize</b> , University of Cambridge                  | 2024 |
| <b>DAAD Scholarship</b> , Deutscher Akademischer Austauschdienst | 2023 |
| <b>Wolfson College Scholarship</b> , Wolfson College, Cambridge  | 2023 |

#### TEACHING

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|-------------------------------------------|------|
| <b>Classical Mechanics 8.01</b> (TA), MIT | Fall |
| <b>Classical Mechanics 8.01</b> (TA), MIT | Fall |

#### SERVICE AND LEADERSHIP

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|------------------------------------------------------------------------------|--------------|
| <b>Treasurer</b> , Edgerton House, MIT                                       | 2025–2026    |
| <b>President MIT Rowing Club</b> , MIT Rowing Club                           | 2026–present |
| <b>Organiser, First Year Seminar</b> , Massachusetts Institute of Technology | 2024–2025    |
| <b>Buddy for first-year students</b> , Heidelberg University                 | 2021–2022    |

#### TECHNICAL SKILLS

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

## LANGUAGES

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German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).

*References available on request.*